

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *MILLENNIUM TIME*, *REDOUBT* AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Millennium Time</i>	<i>Redoubt</i>
Flag	UK	UK
Classification society	N/A	N/A
IMO number	N/A	N/A
Type	Passenger vessel	Tug
Registered owner	City Cruises Plc	Cory Environmental Ltd
Manager(s)	City Cruises Plc	Cory Environmental Ltd
Construction	Steel	Steel
Length overall	31.58m	22.65m
Registered length	N/A	N/A
Gross tonnage	270	86.69
Minimum safe manning	5	N/A
Authorised cargo	N/A	N/A
VOYAGE PARTICULARS		
Port of departure	London (Waterloo Pier)	London (Belvedere)
Port of arrival	London (Tower Pier)	London (Vauxhall)
Type of voyage	Passenger service (internal waters)	Towage (internal waters)
Cargo information	N/A	Towing 3 barges
Manning	5	6
MARINE CASUALTY INFORMATION		
Date and time	17 July 2014 at 1414 (UTC+1)	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	Kings Reach, River Thames, London	
Place on board	Bow	Port side and tow
Injuries/fatalities	None	None
Damage/environmental impact	Structural damage to port bow. Broken windows at the forward end of the passenger saloon	Indented bulwark and deck penetrations. Two towed barges were also damaged.
Ship operation	On passage	On passage
Voyage segment	Transit	Transit
External & internal environment	Fine with good visibility	
Persons on board	362 passengers, 5 crew	6 crew, 1 passenger

1.2 NARRATIVE

At 1407 on 17 July 2014, the passenger vessel *Millennium Time* departed from Waterloo Pier on the south bank of the River Thames (**Figure 1**) with 362 passengers on board. The passengers were a mix of nationalities and included groups of schoolchildren. The vessel's destination was Tower Pier and the vessel's mate was at the helm.

Reproduced from Admiralty Chart BA 3319-0 by permission of the Controller of HMSO and the UK Hydrographic Office.

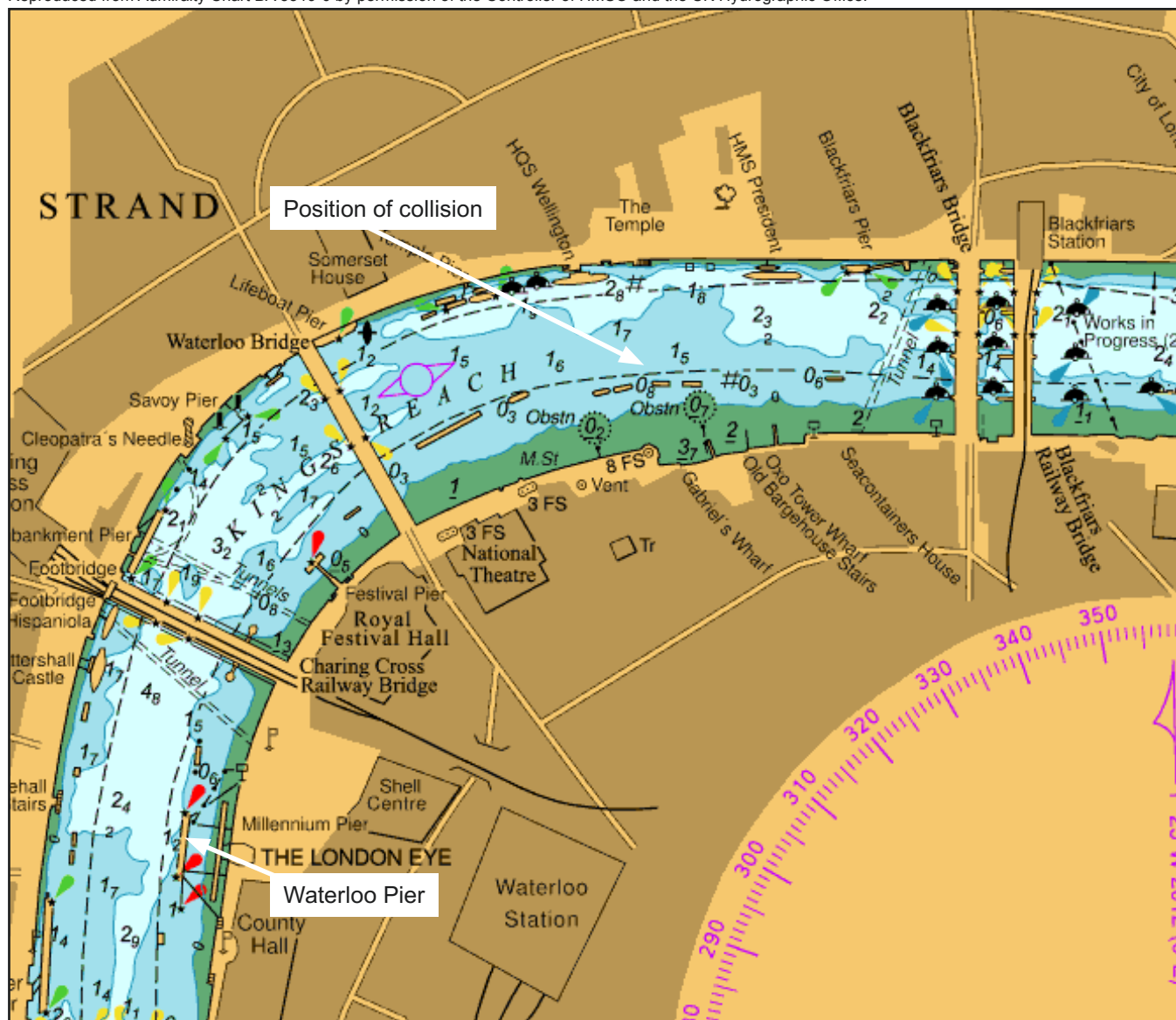


Figure 1: The River Thames from Waterloo Pier to Blackfriars Road Bridge

Millennium Time followed *Mercia*, another passenger vessel, downriver and under Waterloo Bridge. *Monsoon Clipper*, a high speed catamaran on a commuter service, was close off its port quarter (**Figure 2**). *Millennium Time* was making good a speed of 6.6 knots, about 1 knot slower than *Mercia*¹.

The passenger vessels passed under Waterloo Bridge at 1411. *Mercia*'s heading was then adjusted to starboard to follow the axis of the river to the east towards Blackfriars Road Bridge. *Millennium Time* followed approximately 40m astern.

¹ All speeds in this report are speed over the ground.

Meanwhile, *Monsoon Clipper* increased speed to over 12 knots and headed towards Blackfriars Pier on the north bank (**Figure 3**), its next scheduled stop. By now, *Millennium Time*'s skipper was providing a light-hearted sightseeing commentary using the vessel's public address (PA) system. He was standing in the wheelhouse doorway facing aft, towards the passengers (**Figure 4**).

As soon as *Millennium Time* was heading towards Blackfriars Road Bridge, the mate saw that the isophase light on the bridge was lit². The mate was aware that the light probably indicated that a tug and tow were approaching the bridge from the other side. Both *Mercia* and *Millennium Time* were now on a heading of approximately 085°, intending to pass under number 4 arch of Blackfriars Road Bridge³ (**Figure 3**) while keeping a safe distance off the barges on the Coin Street Moorings towards the south bank (**Figure 5**).

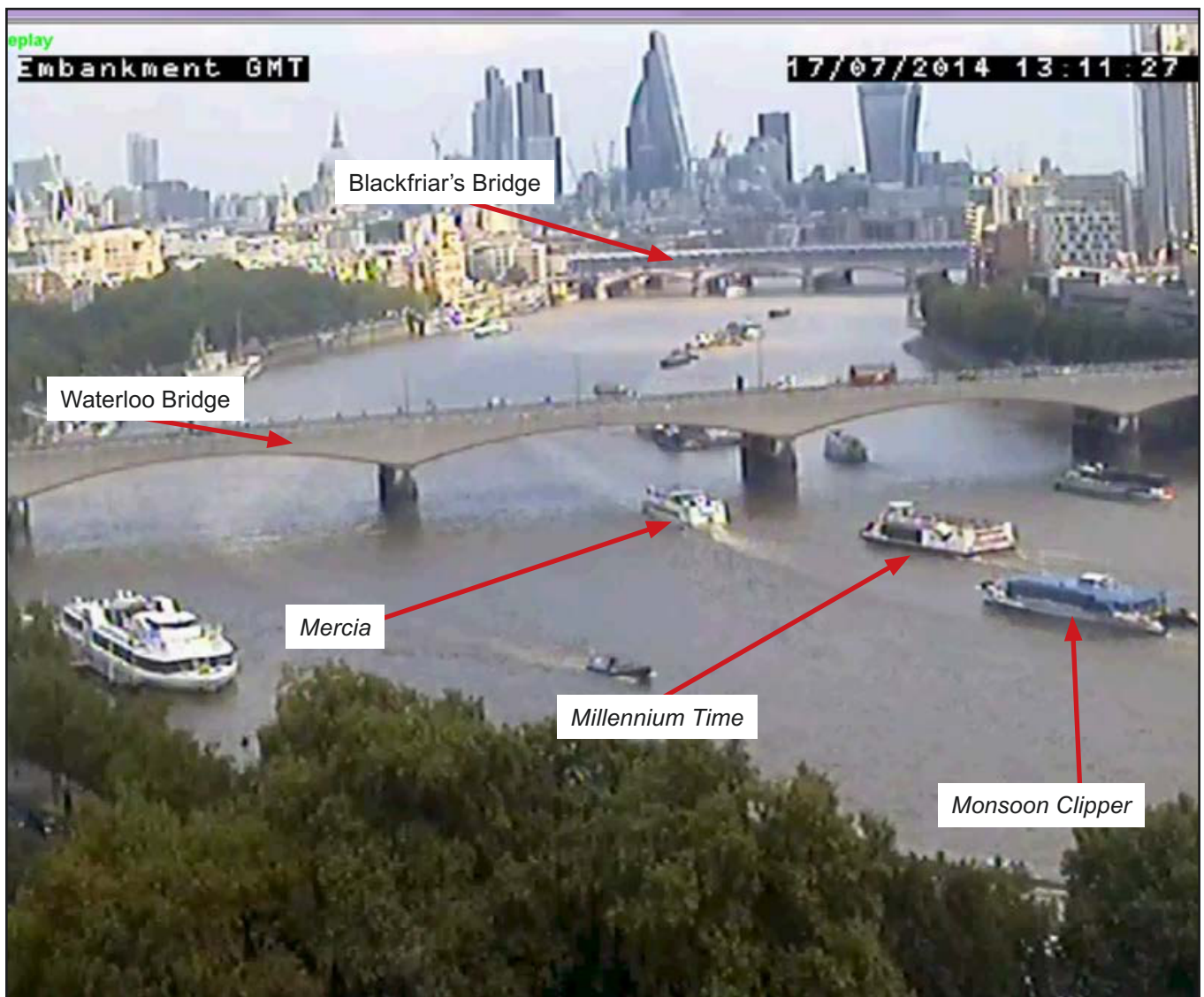


Figure 2: *Millennium Time*, *Mercia*, and *Monsoon Clipper* approaching Waterloo Bridge

² Between Tower and Wandsworth bridges, the bridges on the Thames are fitted with high-intensity white signal lights on both sides of the main navigational arches. The lights are triggered automatically at a distance of approximately 750m by a device fitted to reporting vessels to warn other river users of their approach. An isophase light (2 seconds on, 2 seconds off), indicates that a single reporting vessel is about to navigate under the bridge. A quick flashing light indicates that two or more reporting vessels are approaching.

³ Bridge arches on the Thames are numbered consecutively from the north bank.

At 1413, the motor tug *Redoubt* with three Cringle⁴ barges on a very short tow passed under number 3 arch of Blackfriars Road Bridge (**Figure 3**). The tug and tow were on passage upriver at a speed of 10 knots. The skipper was at the helm and he was accompanied in the wheelhouse by the vessel's five crew.

Immediately after clearing Blackfriars Road Bridge, *Redoubt*'s skipper adjusted the tug's heading to approximately 265° to ensure that the towed barges astern would clear the arch abutments on Waterloo Bridge, the next bridge ahead. He intended to pass to the north of *Mercia* and *Millennium Time*, which he could see approaching off the tug's port bow.

Redoubt and its tow were seen by *Millennium Time*'s mate and its skipper. The skipper checked with the mate that he had seen the tug and then continued with the sightseeing commentary.

At 1414:02, when the distance between *Millennium Time* and *Redoubt* was approximately 110m, *Millennium Time* suddenly turned quickly to port towards *Redoubt*. Over the next few seconds, *Millennium Time*'s mate assumed that the tug had turned towards the passenger vessel and did not take any action. He then turned the helm hard over to starboard and moved the starboard engine control lever to full 'astern'. At about the same time, *Millennium Time*'s skipper saw out of the corner of his eye that the passenger vessel was about to collide with the tug and shouted to the passengers "stand fast". He then threw himself onto the deck.

By this time, *Redoubt*'s skipper had moved the tug's helm hard over to starboard to try and avoid *Millennium Time* but, at 1414.16, the two vessels collided (**Figure 6**). On impact, the tug's skipper put the helm hard to port to try and swing the towed barges clear. However, the passenger vessel struck the port swim⁵ of the southern barge in the head rank. The bow of *Millennium Time* was forced downwards and the barge rode up and hit the passenger vessel's forward saloon windows. The impact parted several lines securing the barges; it also stopped *Millennium Time* in the water. The passenger vessel was then briefly pushed astern at a speed of almost 5 knots.

1.3 RESPONSE AND RESCUE

Immediately following the collision, *Redoubt*'s skipper broadcast a "Pan Pan" urgency message on Very High Frequency (VHF radio). Extracts of the initial radio exchanges between the skipper and London Vessel Traffic Services (VTS)⁶ are at **Table 1**.

⁴ Cringle barges are named after Cringle dock near Battersea in London. The barges are flat bottomed and are designed for carrying containers. The three barges towed by *Redoubt* were arranged in two ranks (two barges in the head rank and one in the rear).

⁵ The 'swim' is the raked square end (bow or stern) of a barge that overhangs the water.

⁶ London VTS manages and oversees the safety of navigation in the Thames between Teddington to the mouth of the river's estuary. It operates from control centres in Gravesend and at the Thames Barrier Navigation Centre in Woolwich.

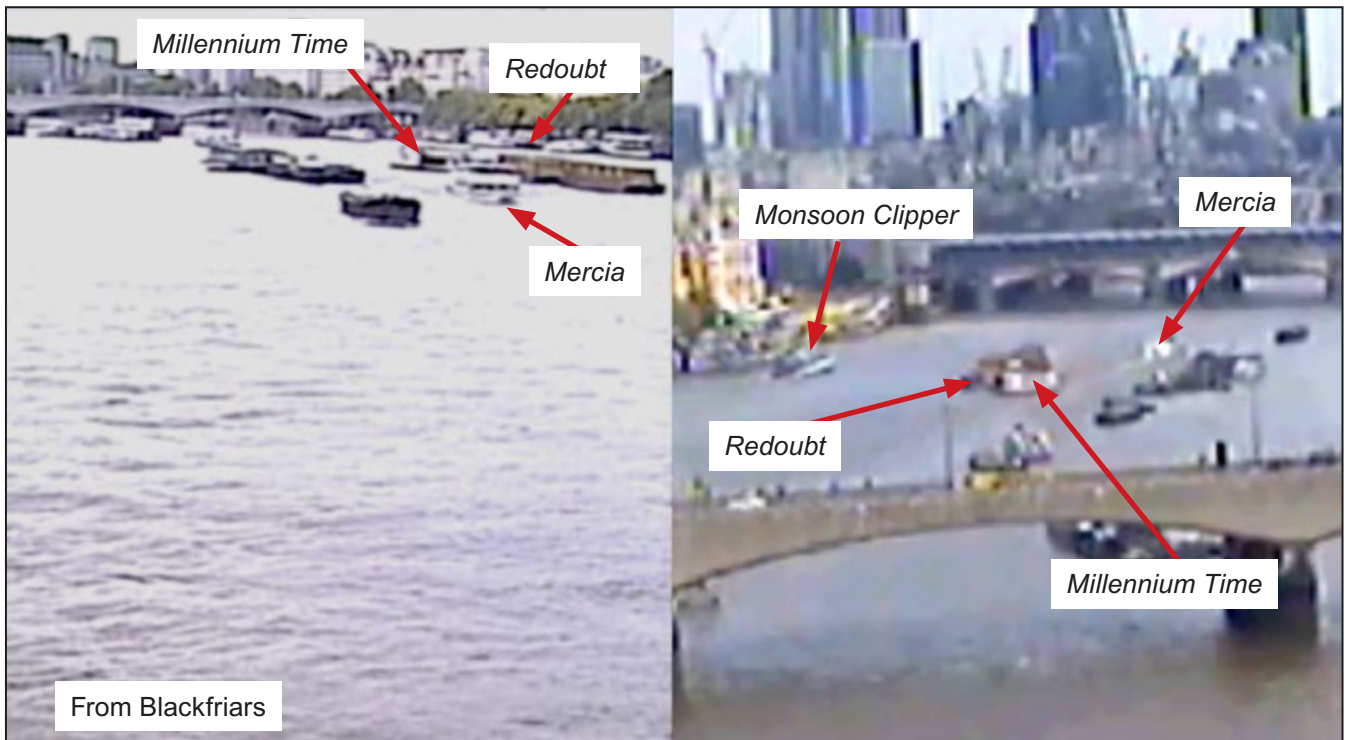


Figure 6: The collision

Time	Transmission by	Transcript
1415:01	Redoubt	<i>London VTS, this is the Redoubt. The Millennium Time has just hit our craft head-on. We are between Blackfriars and Waterloo Bridges.</i>
1415:30	London VTS	<i>Tug Redoubt, London VTS, Roger, can you give us an update; persons in the water, is the vessel sinking?</i>
1415:40	Redoubt	<i>Millennium Time is badly damaged, there has got to be some injury on board. There is no injury on board the tug. I would speak to the Millennium Time or get someone to the Millennium Time; there is probably a lot of injury on board there.</i>

Table 1: Transcript of communications between *Redoubt* and London VTS

At 1417, London Coastguard tasked a Royal National Lifeboat Institution (RNLI) lifeboat from the nearby Tower RNLI lifeboat station to attend the scene. The lifeboat was on site at about 1422 and several of its crew boarded *Millennium Time*. About 3 minutes later, patrol boats with officers on board from the Marine Policing Unit (MPU) arrived to assist. A fireboat from the London Fire Brigade and a Port of London Authority (PLA) harbour service launch with a harbourmaster on board also attended.

The RNLI and MPU noted that the passengers on board *Millennium Time* appeared to be confused and had not been prepared for evacuation. One of the customer service agents appeared to be more focused on clearing the bar area rather than dealing with the needs of the passengers. Also, the crew had not physically checked the extent of the damage to their vessel.

The police took charge and the RNLI crew quickly ascertained there were no major injuries; nine passengers had sustained cuts and bruises and one passenger was suffering with abdominal pains. MPU officers also boarded *Redoubt*.

Millennium Time's crew were unsure of the number of passengers on board. Automatic Identification System (AIS) data transmitted from the vessel⁷ indicated that 426 passengers were on board; the police counted 362. The passengers were transferred to the passenger vessel *Mayflower Garden* and were disembarked at Festival Pier where police and the London Ambulance Service were waiting.

The crews of *Millennium Time* and *Redoubt* were breathalysed for alcohol by the MPU and their mobile telephones were checked to see if they were in use at the time of the collision. The breathalyser tests were negative and no mobile telephones had been in use by the crew immediately prior to the accident.

1.4 ENVIRONMENT

The weather was fine and clear with good visibility. The wind was a gentle breeze (Beaufort force 3) from the east. Low water at Tower Bridge was 1244, 4 days after the spring tide and the height of tide was 4.2m. The predicted tidal stream was flooding at a rate of 2 knots.

A recent hydrographic survey of the river bed in the King's Reach indicated that there had been no significant changes to charted depths since previous surveys. The findings of the survey supported the published data that showed the river-bed to have a relatively flat profile (**Figure 7**). There were no outfalls or drain culverts in the area.

1.5 DAMAGE

Millennium Time sustained structural damage to its bow, and windows at the forward end of the passenger saloon were broken (**Figure 8**). The vessel was taken out of the water and took 32 days to repair.

The bulwarks on *Redoubt*'s port shoulder were pushed inboard (**Figure 9**), which resulted in four penetrations through the main deck. The repair included the removal and replacement of 5m of bulwark plating and capping. *Redoubt* was out of service for 10 days.

Damage to two of the towed cringle barges necessitated the replacement of sections of hull plating.

⁷ *Millennium Time* was one of a number of vessels that are required to carry and operate Thames AIS or AIS A when operating above Victoria Deep Water Terminal.

would have afforded both *Redoubt* and *Millennium Time* more time in which to take evasive action, but that it is unlikely that the collision would have been avoided. The investigation report's recommendations are at **Annex C**.

1.12 THE PORT OF LONDON

1.12.1 The River Thames

The River Thames is the longest river in England and passes through London. It is tidal from its estuary to Teddington Lock and is crossed by 29 bridges along this stretch. The river from Lambeth Bridge to Wapping Ness, in which *Millennium Time* and its sister vessels operated, is recognised to be navigationally complex with strong tidal streams, 11 bridges and a multitude of piers and moorings.

Approximately 40 passenger boat operators trade on the River Thames, resulting in around 200,000 passenger vessel movements carrying 8 million passengers each year; about 750,000 people a year commute to work using the regular shuttle services. Over 70 pleasure vessel clubs and marinas are located on the River Thames, in addition to a large number of rowing clubs. The PLA approves and monitors about 300 organised events on the river each year.

1.12.2 Port of London Authority

The PLA was established in 1909 by the Port of London Act and is the statutory harbour authority for the 95 mile tidal stretch of the River Thames. The PLA's area of responsibility is divided into two districts: Upper (from Teddington Lock to Crossness) and Lower (from Crossness to the Thames Estuary). Each of the districts is administered by a harbourmaster, both of whom report to a chief harbourmaster.

Specific requirements for commercial and leisure craft on the River Thames are detailed in the Thames Byelaws 2012. However, the PLA is empowered by the Port of London Act to issue navigational instructions to masters of vessels through General Directions¹⁸. General Directions issued by the PLA included the requirements for the carriage and use of AIS, vessel reporting and the use of the Central London Special Signal ("isophase") lights. The PLA also issues Notice to Mariners (NTM) to provide information and advice on works and events as well as changes to rules and regulations that might affect river users. NTMs may be either temporary or permanent.

1.13 REGULATION AND GUIDANCE

1.13.1 Construction standards

As a Class V passenger vessel, *Millennium Time* was required to comply with the construction standards detailed in Merchant Shipping Notice (MSN) 1699 -*The Merchant Shipping (Passenger Ship Construction: Ships of Classes III to VI(A)) Regulations 1998*. MSN 1699 does not require ships fitted with non-powered steering to have the rudder angle indicated at the steering position.

¹⁸ A vessel's master may choose not to comply with a General Direction if he can prove that he has reasonable grounds to believe that compliance would imperil his vessel or that compliance was impractical.

MSN 1823 (M) Safety Code for Passenger Ships Operating Solely in UK Categorised Waters, which was published in April 2010, introduced revised regulations for new passenger ships. The code requires that any passenger vessel operating in Category C and D waters is fitted with a rudder angle indicator.

In his Interim Report of the Thames Safety Inquiry¹⁹, Lord Justice Clarke recommends that new safety regulations should, as a general rule, be applied equally to new and existing vessels, subject to a power to grant exemptions based on practicability and the provision of satisfactory alternatives.

In February 2002, in response to Lord Justice Clark's recommendation, the MCA published MGN 218(M) Thames Safety Inquiry Recommendation 27.2 standards for new and existing ships, which stated that in principle:

- (a) All future domestic safety legislation proposed by the MCA should apply equal safety standards to new and existing vessels;
- (b) The legislation should include a power of exemption;
- (c) Exemptions will be issued where compliance with a new standard would be unreasonable (whether on the grounds of practicability or for some other reason), and where an equivalent level of safety is achieved by some means other than strict compliance with the regulations' requirements.

MGN 218(M) also states:

This policy will not be applied retrospectively – ie existing legislation will not be revised in order to conform to this principle. However, where domestic regulations are being revised for other purposes, the principle of a single set of standards should be taken into account.

When MSN 1823 came into force in May 2010, it was the MCA's intention also to introduce legislation in October of the same year restricting the 'grandfather rights'²⁰ of existing domestic passenger vessels. Proposed measures included treating existing vessels as 'new' for the purpose of applying standards if the vessel has undergone major modifications or when a vessel has not been issued with a passenger vessel safety certificate for 5 years or more. Due to changes in government priorities, the intended legislation regarding 'grandfather rights' was not introduced as planned. However, the MCA has continued to develop measures intended to restrict 'grandfather rights' in safety-critical areas such as intact and damage stability, buoyancy, fire prevention, lifesaving apparatus and wheelhouse visibility. The MCA is striving to introduce legislation incorporating these measures within the next 2 years.

1.13.2 Manning and training

Lord Justice Clarke also recommended a review of manning requirements for passenger vessels operating on the Thames to consider whether they were sufficient to ensure the safety of passengers in the event of an emergency. As a

¹⁹ In the early hours of 20 August 1989, the dredger *Bowbelle* collided with the passenger vessel *Marchioness* in Kings Reach on the River Thames. Of the 131 passengers on board *Marchioness*, 51 perished. In 1999, a public inquiry, known as the Thames Safety Inquiry into the *Marchioness/Bowbelle* disaster, was chaired by Lord Justice Clarke.

²⁰ The term 'grandfather rights' refers to the principle that vessels should only be required to comply with the standards that applied when they were built or entered service.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

The collision between *Millennium Time* and *Redoubt* resulted from the passenger vessel turning to port towards the tug and its towed barges. The vessels were only 110m apart and were closing at a speed of over 16 knots on near reciprocal headings. AIS and CCTV evidence (**Figures 5 and 6**) indicates that *Millennium Time* turned 27° in 14 seconds before hitting *Redoubt*.

Prior to the turn, *Millennium Time*'s mate, who was at the helm, was aware of the proximity of the tug, which was due to pass down the passenger vessel's port side at a distance of approximately 15m. The mate did not deliberately initiate the turn by turning the steering wheel to port or by adjusting the engine control levers. Therefore, the turn or sheer must have resulted from several degrees of port rudder being set without the mate being aware, and/or the effects of hydrodynamic interaction. There was no evidence of any mechanical failure of the vessel's steering or propulsion systems.

Millennium Time's mate initially assessed that the tug was turning towards the passenger vessel and it took him several seconds to realise it was in fact *Millennium Time* that was turning. By then, although both the mate and *Redoubt*'s skipper took avoiding action, their efforts were too late to be fully effective.

2.3 RUDDER INDICATION

The absence of rudder angle indication in *Millennium Time*'s wheelhouse meant that the mate had no way of knowing the angle of the rudders. Consequently, it is possible that several degrees of port rudder had been applied when *Millennium Time* unexpectedly turned towards *Redoubt*.

The trials conducted in August 2014 highlighted the difficulty encountered in maintaining *Millennium Time* on a straight course. However, this task was made easier during the later trials in September 2014, for which a rudder angle indicator had been fitted. Although more visual references were also available in the latter trial that allowed the helmsman to steer by 'line of sight', the indicator enabled him to immediately check the angle of the rudders at all times. This removed the need for the helmsman to almost continually monitor the swing of the vessel's stern.

Millennium Time entered service in 1999 and complied with *The Merchant Shipping (Passenger Ship Construction: Ships of Classes III to VI(A)) Regulations 1998* (MSN 1699), which did not require rudder indicators to be fitted on Class V passenger vessels with non-powered steering. The revised regulations for domestic passenger vessels, introduced in MSN 1823 (M) in 2010, applied only to new vessels. Consequently, although new Class V passenger vessels operating in Category C and D waters were required to have a rudder angle indicator, existing vessels such as the M2 class were not.

Nonetheless, the benefits of the rudder angle indicator to *Millennium Time's* helmsman observed during the trials in September 2014, and the circumstances of this accident, support the case for the retro-fitting of rudder angle indication in the wheelhouses of other Class V passenger vessels. This could be done on a case by case basis, taking into account factors such as the steering arrangement, the size of the vessel and the number of passengers carried, the cost of installation and the area of operation.

2.4 INTERACTION

It is possible that *Millennium Time's* sudden sheer towards *Redoubt* was influenced to some extent by the effects of hydrodynamic interaction. The manoeuvring trials conducted in King's Reach in September 2014 showed the extent to which *Millennium Time* was potentially affected by another passenger vessel in close proximity. The passage of the vessel down *Millennium Time's* port side at a distance of about 15m initiated a change of heading of 21° in 20 seconds without any change to the rudder position (**Figure 16**). Although slower than the rate experienced immediately before the collision, this was still very significant.

Before colliding with *Redoubt*, *Millennium Time* was making good about 6.6 knots over the ground against a 2 knot tidal stream and had an under keel clearance of approximately 4.4m. As the under keel clearance was three times the vessel's draught, neither the vessel's speed nor the depth of water are likely to have adversely influenced its movement. In addition, *Millennium Time* did not experience any discernible effects of interaction when passing the Coin Street moorings during the subsequent manoeuvring trials.

Nonetheless, at the time of the sheer which led to the collision with *Redoubt*, *Millennium Time* was passing the Coin Street moorings at a distance of about 15m, *Mercia* was 90m ahead, *Redoubt* was off the port bow at a distance of 110m and *Monsoon Clipper* was overtaking a similar distance off *Millennium Time's* port beam (**Figure 6**). Therefore, this was a complex situation in terms of the hydrodynamic forces at work and it is not possible to single out a particular influencing factor. It is also very difficult to accurately quantify how the potential sources of interaction affected *Millennium Time* given the different hull forms and corresponding pressure zones of the vessels involved.

The width of the navigable water in the King's Reach is significantly reduced by the Coin Street moorings on the south bank (**Figures 2, 3, 5, 6 and 7**). Consequently, vessels frequently have to pass at close quarters and the risk of interaction is increased in this area.

In this case, AIS data indicates that *Redoubt's* track was approximately 15m further south of the tracks historically taken by similar vessels after passing Blackfriars Road Bridge. By itself, this was not unsafe, but it left only about 30m of water between the tug and the moorings for *Mercia* and *Millennium Time* to navigate. Both of the passenger vessels were required by byelaw to avoid impeding the tug and were intending to pass under number 4 arch of Blackfriars Road Bridge. Consequently, the vessels possibly passed closer to the moorings and/or the tug than would otherwise have been necessary.

2.5 REACTION TO THE TURN

2.5.1 *Millennium Time*

As *Millennium Time* suddenly and unexpectedly turned towards *Redoubt*, the mate's perception was that the tug was turning towards the passenger vessel. As a result, he delayed taking any action for several seconds. Given that the interval between the start of *Millennium Time*'s turn to port and the collision was only 14 seconds, this delay was very significant.

As *Millennium Time* turned towards *Redoubt*, the master was facing the passengers on the top deck while providing a commentary. He was aware that *Redoubt* would be passing down the port side but he did not realise that *Millennium Time* was turning towards the tug until just before the vessels collided. Therefore, the commentary clearly distracted the master from his supervision of the mate. If the master had been at the helm, as required by company instructions, or even in the wheelhouse, the probability of *Millennium Time*'s unintended turn being detected in time for effective avoiding action to be taken would have been increased.

During the manoeuvring trials *Millennium Time* responded well to the helm and the interaction-induced sheer in September was easily stopped by the immediate use of counter-helm. Therefore, had the mate intervened sooner, the collision could probably have been prevented. By the time the mate took any action, the vessels were too close for it to be effective.

Although 69 years old, the mate was later proven to be medically fit and there is no evidence to indicate that he was distracted or fatigued. Therefore, his misinterpretation of the relative movement of *Millennium Time* and *Redoubt* was most likely influenced by his knowledge that he had not consciously initiated the turn using the steering wheel. In effect, *Millennium Time*'s turn did not fit with the mate's mental picture of events.

It is also likely that the wheelhouse ergonomics contributed to some degree to the mate's poor situational awareness. There was no rudder angle indicator so he could not easily check whether or not the rudder was amidships. Further, *Millennium Time*'s wheelhouse was positioned forward and had no visual references in the foreground, other than the top of the flag staff that was very close to and below the wheelhouse windows (**Figure 11**). Consequently it would have been harder for the mate to detect *Millennium Time* starting to swing if he was focusing on distant objects, such as the vessels ahead or number 4 arch of Blackfriars Road Bridge, at that time.

The magnetic compass, which would have indicated *Millennium Time*'s change in heading, was not working. However, as most navigation in this area is by 'line of sight' or 'by eye', it is recognised that a compass would rarely be used, if at all. This possibly explains why the compass had not been repaired or replaced and highlights that the defective compass probably made no difference to the mate's response.

2.5.2 Redoubt

Redoubt's master took action to avoid collision by putting the tug's helm hard to starboard as soon as he saw that *Millennium Time* was turning towards his vessel. Although this did not prevent the collision, it possibly prevented the vessels hitting head on, and therefore lessened the severity of the impact. The master's call to London VTS following the collision was also very prompt.

2.6 STEERING SYSTEM

In addition to the problems caused by the lack of any rudder angle indication in *Millennium Time*'s wheelhouse, it is evident from the problems associated with the rod and gear system itself that the system's performance was not sufficient for use on the congested and navigationally challenging waters of the Thames. The manoeuvring trials showed that *Millennium Time* responded quickly when more than 10° of rudder was applied. However, as the rudders were not balanced²³, the force needed to turn the steering wheel when underway at normal operating speed, the number of turns of the steering wheel required to move the rudders through their operating range, and the 'play' between the steering wheel and the rudder, all combined to reduce the system's speed of response and its ease of use.

The rod and gear system fitted to the M2 class vessels had many components and, due to its age, some spare parts were made to order. Given the complexity and overall length of the system with its numerous bevel gearboxes and universal joints, it was inevitable that the maintenance effort required increased over time. Notwithstanding the ongoing replacement of defective or worn parts, the system's performance would have deteriorated since its initial installation and is likely to continue to reduce further. The advantages of changing the rod and gear systems on board these vessels to powered steering was recognised by City Cruises in 2009. Although the twin-screw M2 vessels remained more manoeuvrable than many single-screw vessels operating on the River Thames, it is unfortunate that the plan to fit replacement hydraulic systems was not approved by the City Cruises' Board until 2014, and that none of the vessels had been modified prior to this accident occurring.

2.7 HELMSMAN CERTIFICATION

MSN 1808(M) and Thames byelaws required that the master of *Millennium Time* held a BML with a local knowledge endorsement. None of the vessel's other crew, including the mate, were required to hold a BML and there were no regulatory restrictions on them taking the helm. However, the operational memoranda issued by City Cruises in 2012, in response to accidents involving its vessels and recommendations from the PLA (Section 1.7.2), required a BML licence holder to be at the helm in busy parts of the river, and unqualified helmsmen to be properly supervised by the master in other areas. Although *Millennium Time*'s mate had extensive experience of the Thames and had worked on board the M2 class vessels for 14 years, he did not hold a BML and there is no record of him having completed City Cruises' internal boat-handling assessment. That the mate was at the helm at the time of *Millennium Time*'s collision while the master was providing a commentary to the passengers, shows that these company instructions were ignored.

²³ A 'balanced rudder' is a rudder on which the axis of rotation is behind its forward edge. This minimises the effort required to turn the rudder.

As three of *Millennium Time*'s four scheduled stops (**Annex A**) were between Cherry Garden Pier and Westminster Bridge, Operational Memoranda 11/2012 and 12/2012 had the effect of requiring the master to be at the helm for much of his 9 hour shift. The master would also have had to remain in the wheelhouse for most of the remaining time in order to properly supervise the mate when the vessel was between Cherry Garden Pier and Greenwich Pier. Although not impossible, such a routine would have been difficult to follow. Moreover, the potential additional disadvantage of lost revenue from the sightseeing commentaries made the instructions contained in the operational memoranda, although well-intended, unlikely to be followed.

The traffic density, reduced passing distances, bridges, moorings, strong tidal streams and the likelihood of interaction on the Thames, make it a very challenging area in which to navigate. The policy adopted by some operators to ensure that crew, other than a vessel's master, hold a BML, appears to be a sensible measure to help meet these challenges. Indeed, PLA's recommendation to the MCA in 2012 indicates that the port authority would welcome a regulatory requirement in this respect. However, it would be impractical to require all helmsmen on the Thames to hold a BML, otherwise there would be little scope for training newcomers.

Nonetheless, the provision of more than one BML holder on each vessel has several benefits that lend support to the recommendation in the PLA's investigation report (**Annex C**) for the authority to:

Amend Port of London Authority General Direction 18 – Persons on the bridge to require either the Master, or a deck officer, fully qualified and competent to take navigational conduct of the vessel, to be at the conning position at all times. [sic]

A vessel's master is at all times responsible for the safe navigation of his vessel. Therefore, until the PLA's recommendation is implemented, it is inappropriate for masters, who are the sole BML holders on board, to routinely engage in entertaining passengers while their vessels are transiting congested waters where high speeds of encounter with other vessels and fixed objects exist.

2.8 CREW RESPONSE

When *Millennium Time*'s master saw that the collision with *Redoubt* was imminent, his shouting of 'stand fast' and then diving to the deck contributed little to the safety of the passengers. In contrast, his securing of the vessel on the Coin Street moorings within 3 minutes of the collision was quick and positive, and facilitated their safe evacuation. However, although the vessel had not listed and no bilge alarms had activated, the master should have ensured that a physical check of the vessel's watertight integrity, as prompted in the collision checklist (**Annex B**), was carried out as soon as possible.

The RNLI and the MPU arrived on board *Millennium Time* within 7 minutes of the collision and almost as soon as the vessel had been moored. Until then, the master and mate were busy manoeuvring and securing the vessel. During this period, some of the 362 passengers were undoubtedly shocked and panicked by what had happened.

Millennium Time's crew of five met the required safe manning level indicated on the vessel's Passenger Vessel Safety Certificate to deal with up to 495 passengers. In addition, the in-house training provided to the customer service agents should also have provided a basic level of safety awareness and customer care. However, it is evident that the crew struggled to manage such a large number of linguistically diverse passengers, including schoolchildren, in what was inevitably a chaotic situation.

In such circumstances, strong leadership was required. Although the customer service agents were trained in the onboard procedures and the location of safety equipment, they were possibly also in a state of shock to some degree. They were also not necessarily equipped with the people skills often required during an emergency. This was acknowledged in the customer service agents' terms of reference (paragraph 1.7.3) which required them to follow the master's instructions. However, *Millennium Time*'s master, although having completed crowd management training, was focused on other matters and his ability to direct the customer service agents was limited.

Masters are almost certain to be pre-occupied and have a high workload during an emergency. Therefore, it is important in the short-term that the remaining crew are able to deal effectively with large numbers of potentially panicked and distressed passengers by themselves. Notwithstanding the training requirements of permanent crew detailed in MGN 203 (M), this will frequently be difficult to achieve given the seasonal nature of the customer service agents' employment. Nonetheless, it is important that customer service agents are properly trained to deal effectively with large numbers of passengers in an emergency situation when the deck officers are attending to other aspects of the emergency. To this end, the provision of crowd management training by City Cruises to its crews, in addition to masters, merits consideration.

2.9 PASSENGER COUNT

The Thames Safety Inquiry highlighted the importance of knowing the number of passengers on board a passenger vessel in the event of an emergency. Therefore, it is of concern that 25 years later the number of passengers reported to be on board *Millennium Time* when this collision occurred was so inaccurate.

The number of passengers counted by the MPU on board *Millennium Time* following the collision with *Redoubt* was 362. This was 64 people fewer than the number transmitted on the vessel's AIS, which indicated that 426 passengers were on board. It is evident that this discrepancy was due to the AIS having not been updated following the disembarkation and embarkation of passengers at Westminster and Waterloo piers. Consequently, the vessel did not meet the *Merchant Shipping (Counting and Registration of Persons on Board Passenger Ships) Regulations 1999*, which requires that an accurate record is held on board and ashore at all times.

Even had the passenger numbers been updated on *Millennium Time* following each of its stops, it is unlikely that the passenger count would have been correct. In common with many operators on the Thames, City Cruises relied on its crew to use a hand-operated tally counter to count the passengers on board its vessels. Counting thousands of passengers on and off over a dozen stops each day using this method is extremely difficult and is prone to error. Notwithstanding the difficulty

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. *Millennium Time's* sheer to port, which led to the collision with *Redoubt*, resulted from several degrees of port rudder being set without the mate being aware, and/or the effects of hydrodynamic interaction. [2.2]
2. *Millennium Time* was not required to have rudder angle indication in the wheelhouse but its absence meant that the mate had no way of knowing the angle of the rudders. [2.3]
3. The benefits of rudder indication in *Millennium Time's* wheelhouse during the trials in September 2014, together with the circumstances of this accident, support the case for the retro-fitting of rudder angle indication in the wheelhouses of other Class V passenger vessels equipped with non-powered steering. [2.3]
4. The risk of hydrodynamic interaction was increased in the vicinity of the Coin Street moorings in the King's Reach where the navigable water available was reduced by *Redoubt* being 15m to the south of its usual track. [2.4]
5. *Millennium Time's* mate's perception that the tug was turning towards the passenger vessel was influenced by his knowledge that he had not consciously initiated the turn using the steering wheel and the ergonomics of the wheelhouse. [2.5.1]
6. *Millennium Time's* master did not effectively supervise the mate on the helm because he was providing a sightseeing commentary to the passengers. [2.5.1]
7. The problems associated with the rod and gear steering system fitted on board *Millennium Time* reduced its effectiveness in the congested and navigationally challenging waters of the Thames. [2.6]
8. The advantages of changing the rod and gear steering systems on board the M2 class vessels was recognised by City Cruises in 2009 but none of the vessels had been modified at the time of this accident. [2.6]
9. As the mate on board *Millennium Time* did not hold a BML, company instructions required the master to be at the helm. These instructions were ignored. [2.7]
10. Operational memoranda regarding the wheelhouse manning on board *Millennium Time* were unlikely to be followed by the master and the mate due to the vessel's schedule and the potential loss of earnings from sightseeing commentaries. [2.7]
11. It was inappropriate for *Millennium Time's* master to be engaged entertaining the passengers, leaving the unqualified mate at the helm. [2.7]
12. The traffic density, reduced passing distances, bridges, moorings, strong tidal streams and the likelihood of interaction on the Thames make it a very challenging area in which to navigate. The policy adopted by some operators to ensure that crew, other than a vessel's master, hold a BML, appears to be a sensible measure to help meet these challenges. [2.7]

13. It is evident from the circumstances of this accident and the accidents involving M2 vessels since 2012 that City Cruises' safety management and the safety culture of its crews have yet to develop fully. [2.10]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. It is possible that *Millennium Time*'s sudden sheer towards *Redoubt* was influenced by the effects of hydrodynamic interaction. However, the hydrodynamic forces at work were complex and it is difficult to accurately quantify the collective affects of their potential sources. [2.4]

3.3 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The failure to physically check the watertight integrity of *Millennium Time* following the collision was a significant omission, and the apparent ineffectiveness of the vessel's crew in managing the passengers is of concern. [2.8]
2. *Millennium Time*'s customer service agents were trained in emergency procedures and the location of safety equipment. However, this training did not fully equip them with the people skills that are often required in chaotic situations during an emergency. [2.8]
3. *Millennium Time* had 64 fewer passengers on board than the number transmitted on the vessel's AIS. The AIS data had not been updated following the disembarkation and embarkation of passengers at Westminster and Waterloo piers. [2.9]
4. Notwithstanding the difficulty for crews to make sure that passengers are not missed or are double counted, the counting of passengers with a hand tally counter is repetitive and monotonous and can potentially lead to crews becoming inattentive and/or easily distracted. [2.9]
5. Not all passenger vessels operating on the Thames are constructed with watertight sub-divisions. Consequently, the potential for a far more serious accident remains and the MCA's intentions to restrict 'grandfather rights' in safety-critical areas should be supported. [2.11]